

Exploiting Brownian Motion of Plasmonic Nanoparticles Using Optical Printing Approach for on-Demand Physical Unclonable Functions

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An on-demand fabrication method for additive physical unclonable functions (PUFs), a hardware-based security primitive, is inevitably required, especially considering increasingly miniaturized microelectronic devices. An optical printing approach is regarded as an alternative method to fabricate functional nano/microscale patterns against conventional methods due to its superior fabrication flexibility. However, owing to the Brownian motion of nanoparticles, achieving highly precise and selective printing persists an ongoing obstacle for the applicability of optical printing methods. Here, it is shown that the optical printing approach possesses plenty of room to fabricate on-demand PUFs by exploiting the obstacle from the perspective of randomness. To demonstrate this, an optical PUF based on a mesoscopic lattice pattern consisting of optically printed gold nanoparticles is proposed. Comprehensive analyses on physical features occurring naturally and multi-modal keys generated from them reveal that both exhibit randomness. Through a ternary bit system and key integration approach, the capability of the physical unclonable function using as few as 25 nanoparticles can be ensured in terms of the amount of information, complexity, uniqueness, and encoding capacity. The versatility of optical printing regarding the usability of a broad range of substrates and the ability to create arbitrary patterns with tunable dimensions are also shown.

1. Introduction

Amidst the massive influx of products and the intricate global supply chain, counterfeit products are on the rise. Particularly, despite being manufactured with state-of-the-art technology, counterfeiting crucial microelectronic components, such as integrated circuits, within a system is considered a serious problem in the present semiconductor industry.^[1–3] This can cause a myriad of issues beyond economic and intellectual property losses of supply companies, including threats to public safety and end users's lives, especially when counterfeit microelectronic devices are employed in automobiles and military systems.^[4,5] To abate this situation, the demand for robust secure identification and authentication methods has recently heightened.^[6]

At present, dominant security systems rely on software-based cryptography.^[7,8] Yet, these digital security systems are susceptible to cryptography attacks (e.g., cryptanalysis attack) exploiting vulnerabilities in the cryptographic systems and mathematical flaws in the algorithms.^[9–11] To

address these problems, hardware-based cryptography utilizing physical unclonable functions (PUFs), which exploit uncontrollable and unpredictable physical variations occurring naturally during the manufacturing process, has emerged as an alternative approach.^[12–15] These inherently random features, analogous to a fingerprint, render PUF replication nearly impossible, allowing it to generate a unique key. Electrical PUFs commonly use unpredictable variation of electrical responses in electronic devices, but they may be susceptible to AI-driven modeling attacks or lack entropy sources.^[16–18] To complement their drawbacks, additive PUFs such as nano/microscale optical patterns can be considered.

To incorporate the additive optical PUFs into increasingly miniaturized microelectronic devices,^[19] the dimensions of both materials and fabrication methods for the PUFs should be comparable with them. Within this context, various optical nanomaterials can be considered candidates, including organic

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materials,^[20,21] quantum dots,^[22,23] 2D materials,^[24–26] and plasmonic nanoparticles (NPs).^[27–30] Among these, plasmonic NPs are garnering attention owing to their unique optical properties that stem from localized surface plasmon resonance (LSPR).^[31] LSPR significantly enhances the light–matter interaction at specific wavelengths, making it feasible to measure the optical response of single plasmonic NPs.^[32] Additionally, the diverse responses from plasmonic NPs, such as absorption, scattering, and near-field enhancement, can facilitate the generation of multi-modal challenge-response pairs. These capabilities enable straightforward authentication of PUFs and can show potential for applications that require PUFs within confined areas. Recently, PUFs utilizing plasmonic NPs have been fabricated using drop-casting,^[27,33] self-assembly,^[34,35] and lithographic methods.^[28,36] Although previous studies have successfully demonstrated the potential of plasmonic NPs as optical signal resources for optical PUFs, these conventional methods lack fabrication flexibility required for on-demand production at the mesoscopic scale.

Optical printing,^[37–39] which is an all-optical method to fabricate patterns and devices by directly immobilizing singular NPs in diffraction-limited region on substrates using a tightly focused laser beam, may be highly suitable for the on-demand fabrication of mesoscopic PUFs. Optical printing techniques provide outstanding fabrication flexibility, compared to conventional top–down (e.g., lithographic techniques) and bottom–up (e.g., self-assembly) methods, in using diverse types of NPs^[40–44] and substrates^[45–47] and fabricating arbitrary patterns and structures with high throughput system.^[48–50] Furthermore, nanometric positional errors and pseudo-selectivity induced by Brownian motion of NPs during optical printing, which remain a fundamental challenge in optical printing, can be exploited as an entropy source to fabricate unique PUFs.

In 1905, Albert Einstein spearheaded the establishment of a theory^[51] concerning Brownian motion, which is a jittering movement of a particle suspended in a fluid, followed by numerous physicists and mathematicians.^[52] The dynamics of Brownian particles over long time scales can be described stochastically through a Langevin equation.^[53] However, accurately measuring and predicting a 3D trajectory of a NP and manipulating the movement in real-time for the highly consistent and selective optical printing is generally challenging. This is because the smooth ballistic motion of NPs in a liquid medium at very short times is induced by continuous and rapid collisions of particles with fluid molecules, which are typically assumed to be random.^[54–56] Furthermore, despite significant advancements in nanomanipulation techniques such as plasmonic nanotweezers,^[57–59] optoelectronic tweezers,^[60] opto-thermophoretic tweezers,^[61,62] and opto-thermoelectric tweezers,^[63] suppressing the Brownian motion and manipulating the NPs with highly precise manner have been an ongoing challenge in the field of nanotechnology since its inception. Therefore, with no silver bullet solution for governing the Brownian motion, an inevitable occurrence of nanometric positional errors and pseudo-selectivity in optical printing can be fundamentally regarded as uncontrollable and unpredictable. However, from a reverse perspective, if the consequent errors induced by the Brownian motion during optical printing exhibit sufficient randomness, exploiting them can offer plenty of room

for optical printing technique to fabricate on-demand mesoscopic PUFs.

Herein, we show randomness of physical features induced by the Brownian motion during optical printing and propose a plasmonic ternary PUF consisting of tens of AuNPs that thoroughly exploits this randomness. Initially, we explored the feasibility of utilizing Brownian motion as a source of randomness for fabricating PUFs by comprehensively analyzing the randomness of the consequent features, which remain underexplored in the field of optical printing. On this basis, we generated multi-modal ternary bit keys from the independent random variables in a PUF and suggest an approach that integrates these individual keys into a single key to create a stronger PUF consisting of a limited number of NPs. Furthermore, considering the confined space, we determined the minimum number of NPs required for the proposed PUF to function properly by conducting systematic analyses on the PUF in terms of amount of information, complexity, uniqueness, and encoding capacity. We also validated the applicability of our PUF for applications such as cryptography and IoT protocols and demonstrated the versatility of optical printing as a highly promising method to fabricate on-demand PUFs for mesoscopic devices.

2. Results and Discussion

2.1. Randomness Induced by Brownian Motion during Optical Printing

The optical setup for fabricating PUFs and obtaining the optical response of the fabricated PUFs is illustrated in **Figure 1a**. A laser at wavelength of 532 nm, highly focused on the substrate through 100× objective lens (NA = 1), is used for optically printing singular AuNPs. A dark-field condenser is utilized to collect only the scattered light from AuNPs, while blocking the transmitted light. The light scattered from single AuNPs can be visualized through a complementary metal-oxide-semiconductor (CMOS) camera, and the corresponding optical scattering spectra can be measured using a spectrograph and an electron-multiplying charge-coupled device (EMCCD) (see details in Experimental Section).

Figure 1b depicts a schematic illustrating the fabrication process of a proposed PUF on a substrate using an optical printing technique. The PUF consists of three types of AuNPs, optically printed in a lattice pattern with predetermined spacing, showing haphazardly distributed types and positions of printed AuNPs.

Figure 1c represents a brief workflow of this study. Initially, we fabricated plasmonic PUFs that exploit the randomness induced by the Brownian motion during optical printing of AuNPs. Rapid and stochastic collisions in three dimensions between AuNPs and surrounding small fluid molecules make the precise replication of printed positions of the AuNPs unattainable (left in **Figure 1c**). After fabrication of the PUFs, multi-modal ternary bit keys were extracted based on the optical responses from individual AuNPs within the PUFs. The lateral positions and types of AuNPs, extracted from optical dark-field images and optical spectral responses under white light illumination, were utilized for key generation (middle in **Figure 1c**). Afterward, the capability of the proposed plasmonic ternary PUFs was validated through statistical analyses and information theory conducted

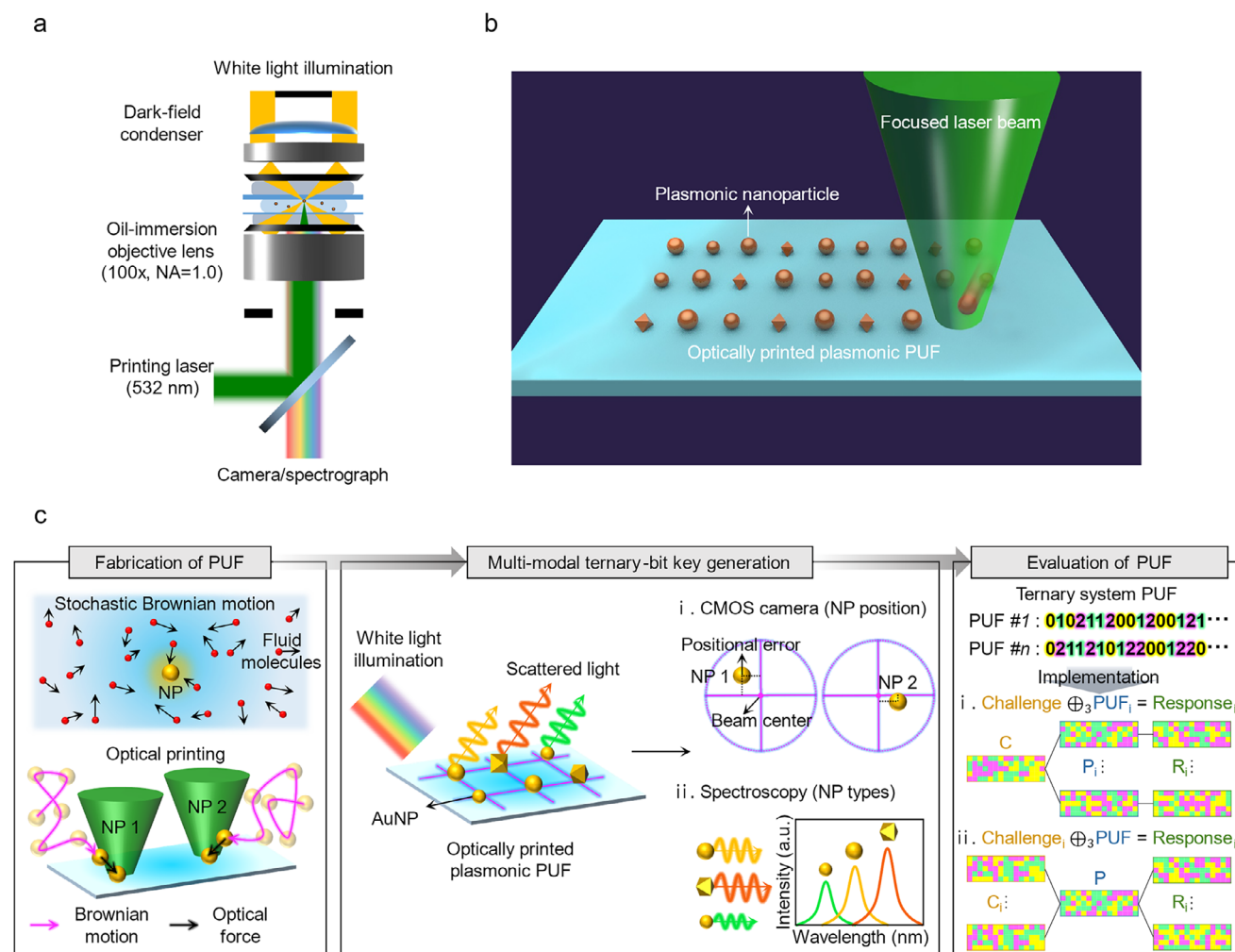


Figure 1. a) Schematic illustration of the optical setup for optical printing of AuNPs and obtaining optical responses from printed AuNPs. b) Schematic illustration of the fabrication process of a proposed plasmonic PUF consisting of AuNPs. c) A brief workflow of this study.

on the ternary keys. Additionally, implementation of the ternary PUFs as a unique security medium was demonstrated based on an exclusive-OR operation (right in Figure 1c).

The process of optical printing involves two steps as shown in Figure 2a. First, a NP driven by Brownian motion enters a region called the “capture volume”^[64,65] of a focused laser beam where optical forces overwhelm the diffusive motion of the NP. Subsequently, the NP experiences the deterministic motion toward an end position to be printed on a substrate driven by the optical forces. If the optical forces exerted on the NP at this end position are sufficient to overcome threshold forces between the NP and the substrate, the NP is immobilized onto the substrate via van der Waals interactions. A range of the end positions where NPs can be immobilized on the substrate is defined as the “printable region” and its size is determined by a relationship between the optical forces exerted on the NPs and the repulsive forces. When the threshold is fixed constant, an increase of laser power leads to an enlargement of the printable region to the laser spot size (see Figure 2b). For instance, Figure 2c displays optical dark-field images of printed 100 nm AuNPs which

were intended to be printed at the vertices of equilateral triangle, with the laser power increasing from 2.5 to 10 mW. As the laser power increases, a transformation from an equilateral triangle to an irregular configuration was observed, indicating that the increase in laser power magnifies the printable region. Theoretically, minimizing optical forces to the threshold by decreasing the laser power gives the impression that the printable region can be reduced until it reaches a point. However, a systematic study^[66] on accuracy and mechanistic details of optical printing of plasmonic NPs revealed that achieving a printable region smaller than tens of nanometers is not feasible because the value of the printable region sharply decreases for realistic decrements in laser power. Furthermore, fluctuations in the repulsive threshold forces and optical forces exerted on colloidal NPs, which exhibit morphological heterogeneities, render this more challenging in practice. The concepts of “capture volume” and “printable region” in optical printing have been extensively elucidated both theoretically and experimentally in previous studies.^[39,66,67] However, there is still a dearth of research analyzing the printed positions of NPs, particularly regarding

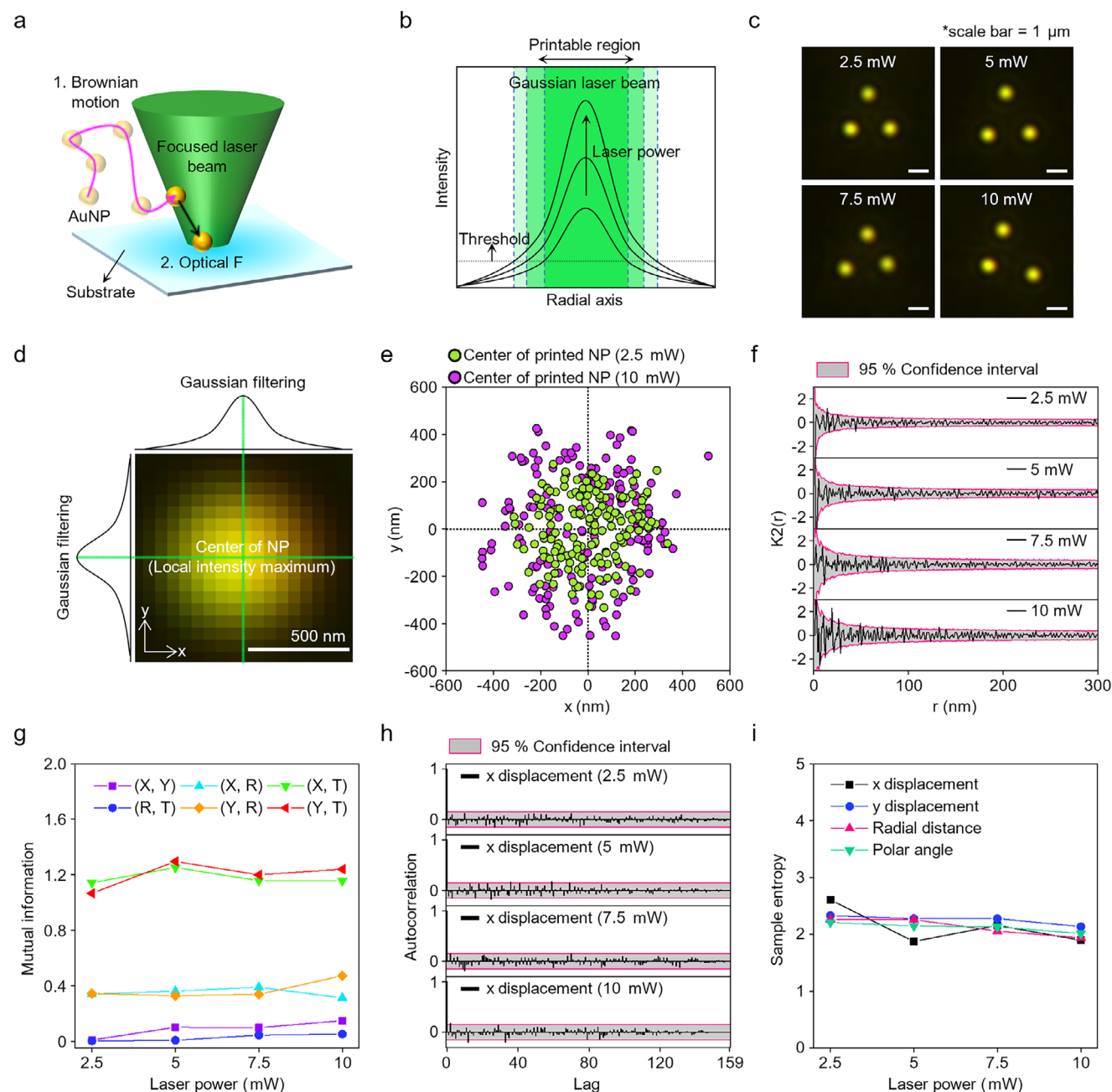


Figure 2. Analyses on randomness induced by Brownian motion during optical printing of AuNPs. a) Schematic illustration of the optical printing process. b) Schematic illustration of the printable region depending on the laser power with a constant threshold. c) Optical dark-field images of optically printed 3 AuNPs on glass substrates. d) A dark-field image of an AuNP to determine the center position using local intensity maxima with Gaussian filtering. e) Scattered patterns displaying the center positions of printed AuNPs (160 each) with laser powers of 2.5 mW (green dots) and 10 mW (purple dots). f) $K2(r)$ of the scattered patterns. Black lines and red lines denote the observed $K2(r)$ and 95% confidence interval of 999 simulated CSR patterns, respectively. g) Mutual information of combinations of x displacement (X), y displacement (Y), radial distance (R), and polar angle (T). h) Autocorrelation function of x displacement. Black lines and red lines represent autocorrelation and a 95% confidence interval, respectively. i) Sample entropy of x displacement, y displacement, radial distance, and polar angle as a function of the printing laser power.

their distribution, such as uniformity, aggregation tendencies, or randomness.

To analyze the power dependency on the distributions, we optically printed 100 nm AuNPs on a glass substrate using laser powers of 2.5, 5, 7.5, and 10 mW. For distribution analysis, cen-

ter position of a printed NP was defined as local intensity maxima using gaussian filtering (see Figure 2d). Figure 2e shows scattered patterns displaying the center positions of AuNPs optically printed with laser powers of 2.5 mW (green dots) and 10 mW (purple dots). The results printed with powers of 5 and

7.5 mW are shown in Figure S1a,b (Supporting Information), respectively. As expected, the AuNPs printed using lower laser power are distributed in a narrower area compared to those of printed with higher laser power due to the smaller printable region (see Figure 2e). To quantitatively compare the distribution areas of AuNPs printed with each laser power, the positional errors, defined as the root mean square deviation in the radial direction, were represented as a function of laser power in Figure S1c (Supporting Information). In the laser power range from 2.5 to 7.5 mW, the positional error tended to increase from ≈ 201 to ≈ 294 nm and then converged to ≈ 300 nm at 10 mW, implying that at laser powers of 7.5 mW and above, the printable region expanded to the spot size of the focused laser beam.

To determine whether the scattered patterns of center positions of printed AuNPs exhibit uniformity, aggregation, or randomness, we used K2 function,^[68] one of the spatial point pattern analysis methods. Although Ripley's K function^[69–71] and pair correlation function (PCF)^[72] are more commonly used to characterize point patterns,^[73] these methods are insufficient for addressing spatial heterogeneity.^[68,74] Therefore, the K2 function, an approximation to the first derivative of the PCF, was utilized in this study. The patterns, as shown in Figure 2e and Figure S1a,b (Supporting Information), with 999 complete spatial randomness (CSR) patterns generated by Monte-Carlo simulations were analyzed using the K2 function (Figure 2f). The black lines in Figure 2f, representing the K2(r) of the sample patterns, fell within the grey area showing the 95% confidence interval of the simulated CSR patterns. This indicates that although the diameter of the patterns is determined by the printing laser power, points in the patterns are randomly distributed regardless of the printing laser power.

Optical printing proceeds sequentially, but the spatial point pattern analysis lacks consideration for the temporal order of pattern formation. Therefore, additional analysis methods for time-series data are required. Prior to conducting the analyses, we decomposed the printed position of AuNPs represented in two-dimensions into four variables according to printing event: x displacement, y displacement, radial distance, and polar angle, as shown in Figure S2 (Supporting Information). And we assessed the interdependence between these variables using mutual information.^[75,76] As depicted in Figure 2g, mutual information of (X (x displacement), Y (y displacement)) and (R (radial distance), T (polar angle)) exhibits low interdependence, whereas mutual information of (X or Y , R or T) shows higher interdependence because R and T are transformed variables from X and Y . These results are also confirmed by the three types of correlation coefficient (see Figure S3, Supporting Information). This indicates that one variable, such as X and R , does not affect the other variable, such as Y and T . Therefore, considering the independence of these variables, we assessed the randomness of printing events by analyzing the time-series variable data as follows.

Autocorrelation function (ACF) is utilized to demonstrate the independence of printing event through measure of autocorrelation of the displacement, radial distance, and polar angle of the AuNPs at each printing event. ACF of x displacements of the printed NPs lies within the 95% confidence interval as shown in Figure 2h, implying that there is lack of periodic patterns of the data. Similar observations are noted for y displacements, radial distances, and polar angles as well (see Figure S4, Supporting In-

formation), indicating that the printed positions of the NPs are independent from one printing event to the next. These results are further substantiated through sample entropy^[77] analysis, as depicted in Figure 2i. The sample entropy values close to 2 and low variance signify the irregularity and complexity of the data series irrespective of the printing laser power. We further assessed the complexity of the data using permutation entropy,^[78] which considers ordinal patterns of the data (see Figure S5, Supporting Information). The permutation entropy values of the variables closely approximate 1 regardless of the power, demonstrating the high complexity of the data.

Based on these comprehensive analyses using statistical methods and information theory, we ascertained that the positions of optically printed AuNPs within the printable region exhibit randomness, possess complexity, and lack periodicity between printing events, making them unpredictable. This can be attributed to the Brownian motion. Within a specified printable region, terminal positions of NPs to be printed are determined by a combination of the initial conditions such as entry position, velocity, and direction upon entering the capture volume and the deterministic motion by optical forces, as previously described. Yet, the Brownian motion causes NPs to stochastically enter the capture volume, and subsequently deterministic optical forces and directions exerted on the NPs fluctuate with each event. This unpredictably results in inevitable positional errors relative to the designated position, leading to the printed NPs being randomly distributed within the printable region. To demonstrate the sensitivity of optical printing on the initial conditions, we performed particle dynamics simulations in which NPs enter the fixed optical field from the same position and with constant velocity, but with different directions (see Figure S6, Note S1, Supporting Information for details). The simulation results reveal entering direction-dependent distinct and diverging terminal positions under deterministic optical forces, demonstrating that the optical printing system exhibits extreme sensitivity to initial conditions. Therefore, our optical printing approach does not rely on Brownian motion alone but leverages a stochastic-to-deterministic transformation process, where Brownian motion-induced minute fluctuations are deterministically mapped into diverse positional outcomes. This mechanism provides a robust and physically grounded entropy source, and the resultant random features experimentally observed can be potentially exploited as a fundamental source of randomness for fabricating unique PUFs.

2.2. Fabrication of Plasmonic PUF

To make PUF stronger, approaches such as vertical stacking of PUFs,^[79] adopting multi-level challenge-response pairs,^[79–81] and employing a multinary bit system^[82–84] for key generation have been developed. Considering PUFs in confined space, the latter may be a more suitable choice. Thus, we used an ink consisting of a mixture of three types of AuNPs (80 and 100 nm AuNPs, and octahedral AuNPs) suspended in water at an equivalent concentration to fabricate a ternary plasmonic PUF. The SEM images of these AuNPs are shown in Figure S7 (Supporting Information). To demonstrate the ternary PUF, a 9×9 lattice pattern was optically printed by systematically moving the focused beam center

on the glass substrate at 3 μm spacing, as illustrated in **Figure 3a**. In **Figure 3b**, dark-field images of lattice patterns printed using laser powers of 5, 7.5, and 10 mW show that 81 AuNPs were deposited to form the lattice. The images in the bottom right corner, which are green, orange, and yellow dot, represent three distinct types of single AuNPs that compose the lattice.

If discernible patterns emerge in the printed positions or types of AuNPs during optical printing of the three types of AuNPs, they can potentially impose constraints when used for PUFs. Therefore, to confirm the tendency of patterns and optimize the printing laser power for PUF fabrication, we first characterized the power dependency on the printed positions of the AuNPs. Three types of AuNPs, 150 of each type, were printed using laser powers of 5, 7.5, and 10 mW, and their center positions are shown in **Figure 3c** and **Figure S8a** (Supporting Information) for each printing power. Blue, green, and orange dots represent 80 nm AuNPs, 100 nm AuNPs, and octahedral AuNPs, respectively. As the laser power increases from 5 to 10 mW, the positions progressively spread out and exhibit overlapping. For quantitative comparison, the positional errors of the three types of AuNPs are plotted against the laser power in **Figure 3d**. The results indicate that at 10 mW, the positional errors of the three AuNPs nearly converge to ≈ 299 , ≈ 300 , and ≈ 306 nm, respectively. Since positional error values provide only partial information regarding their distribution from the laser center, for a more comprehensive comparison, we analyzed x and y displacements using cumulative distribution functions (see **Figure 3e**; **Figure S8b,c**, Supporting Information). It was clearly confirmed that at 5 and 7.5 mW, there are differences in the cumulative probability of the three AuNPs, whereas at 10 mW, the distribution of them is nearly uniform across the entire range. This implies that at 10 mW, the three types of AuNPs can be randomly distributed with equal probability within the printable region, suggesting their potential use as unique key generators.

Additionally, power dependency on printed AuNP types was analyzed based on six sets of 9×9 lattice patterns fabricated for each power (see **Figure 3b**; **Figure S9**, Supporting Information). **Figure 3f** shows the population of printed AuNP types as a function of laser power. The types of AuNPs are classified by peak wavelengths of optical scattering responses obtained from printed AuNPs. **Figure S10a** (Supporting Information) shows the optical scattering spectra of the three types of single AuNPs. Upon plotting the scattering peak wavelengths (see **Figure S10b**, Supporting Information), it is evident that distinguishing the three AuNPs could be done by dividing at 580 and 630 nm. Similar to the results of the printed positions, it was observed that the population converges as the laser power increases to 10 mW (see **Figure 3f**). These results suggest that at lower powers, relatively smaller AuNPs may enter the capture volume faster and be printed first due to particle size-dependent diffusion.^[85] Whereas the uniform population at 10 mW might be attributed to the expanded capture volume and printable region, which are sufficient to accommodate variations in diffusion.

Furthermore, to investigate the periodicity of printed AuNP types, we plotted the AuNP types of six sets of patterns in sequential order against the printing event, as depicted in **Figure S11** (Supporting Information). The data series were analyzed using the ACF, confirming that autocorrelation is within the 95% confidence interval for all three powers (see **Figure 3g**), signifying an

absence of periodicity in the data. Moreover, we used Lempel–Ziv complexity (LZ complexity)^[86] to assess the complexity of the data. The calculated LZ complexity values are comparable to those obtained from 100 000 simulations of datasets where ‘0’, ‘1’, ‘2’ are randomly distributed in the same sequence length (dotted line), indicating high complexity in printed NP types (see **Figure 3h**). These results imply that it is uncertain which type of AuNPs will be printed in the next printing event.

Through the experimental exploration of power dependency and subsequent analysis, it was demonstrated that optically printing three types of AuNPs using a sufficiently high laser power (10 mW in this study) can induce unpredictable randomness in both position and NP type. These characteristics, despite being contrary to the research direction in the optical printing field, which aims to reduce positional errors and achieve selective printing in heterogeneous colloids,^[67] can be effectively exploited in the key generation process for the proposed plasmonic ternary PUF. Additionally, we evaluated the printing throughput by measuring the waiting time required to print a single AuNP (see **Figure S12**, Supporting Information). The median waiting time per each printing event is 5.625 s, demonstrating the practical feasibility of our optical printing approach for PUF fabrication.

2.3. Key Generation and Evaluation of PUF

Using a multinary bit system for key generation and extending the length of the key can create a strong PUF with a limited number of AuNPs. For instance, utilizing multiple discrete states such as ternary and quinary bits, significantly enhances the complexity and encoding capacity, which aligns with the direction of improving PUF robustness through complex encoding schemes.^[79,80,87,88] To achieve this purpose, we initially explored the feasibility of utilizing the entire randomness of printed positions and NP types generated during the optical printing of the three types of AuNPs for PUF keys. To perform this, 20 PUFs were fabricated in a 9×9 lattice pattern utilizing a laser power of 10 mW and three variables from a single AuNP in the pattern, including x position, y position, and NP type, were adopted as potential candidates (see **Figure 4a**). The distance in the x and y axis from the beam center to the center position of the NP, defined as δx and δy , was employed to generate a ternary bit key for the x position and y position, respectively. The criteria for ‘0’, ‘1’, ‘2’ in the ternary bit system were statistically determined based on the cumulative distribution function in **Figure 3e**, using the mean displacement values at cumulative probabilities of 0.333 and 0.666. Accordingly, the x position thresholds are set at 104.08 and 225.1 nm, while for the y position, they are 91.7 and 199 nm, respectively. And the criteria for ‘0’, ‘1’, ‘2’ for NP type were based on the scattering peak wavelengths (see **Figure S10**, Supporting Information). For instance, in the case of AuNP shown in **Figure 4a**, we can extract ‘1’ from the x position, ‘2’ from the y position, and ‘0’ from the NP type.

We implemented this process to generate three types of ternary bit keys for a PUF fabricated in a 9×9 lattice pattern. **Figure 4b** displays a dark-field image of PUF #1. The positional distribution and population of AuNPs comprising this PUF are shown in **Figure 4c,d**, respectively. The consequent ternary bit keys

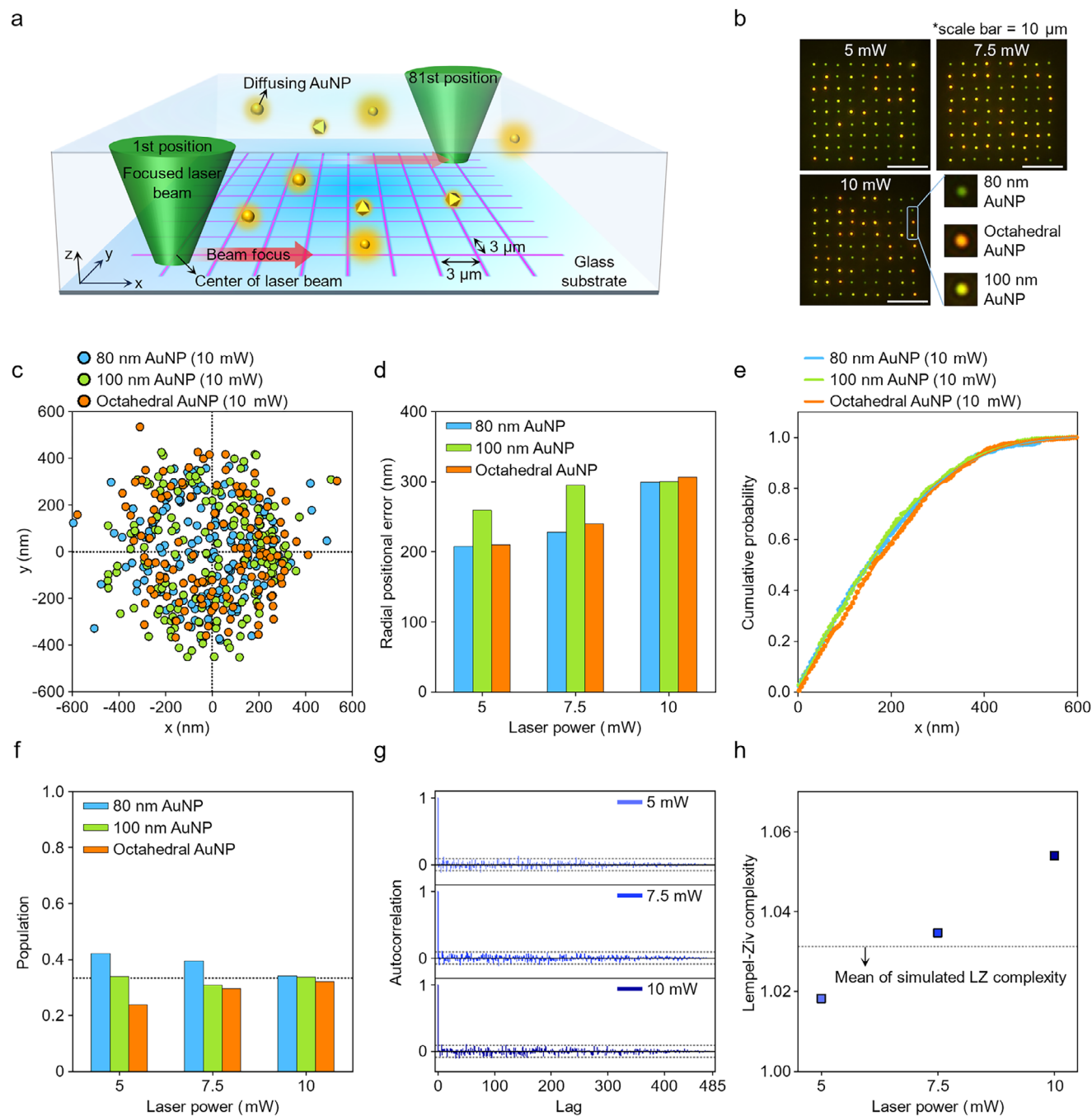


Figure 3. Fabrication of plasmonic PUF using a mixture of three types of AuNPs. a) Schematic illustration of the fabrication process of a 9×9 lattice pattern with a $3 \mu\text{m}$ interval for PUF on a glass substrate. b) Dark-field images of the lattice patterns optically printed using laser powers of 5, 7.5, and 10 mW. The bottom right corner images display three distinct types of single AuNPs (80 nm AuNP, 100 nm AuNP, and octahedral AuNP). c) Scattered patterns showing the center positions of printed three types of AuNPs (150 each) using the printing laser power of 10 mW. d) Radial positional errors of the three types of AuNPs as a function of the laser power. e) Cumulative distribution function of x displacements of three types of AuNPs printed using the laser power of 10 mW. f) The population of printed AuNP types as a function of laser power. The total number of AuNPs printed each laser power is 486. g) Autocorrelation function of printed AuNP types. Dotted lines represent a 95% confidence interval. h) Lempel–Ziv complexity of printed AuNP types. The dotted lines show the mean value of LZ complexity calculated from 100 000 simulations of time-series datasets where '0', '1', '2' are randomly distributed in a sequence length of 486.

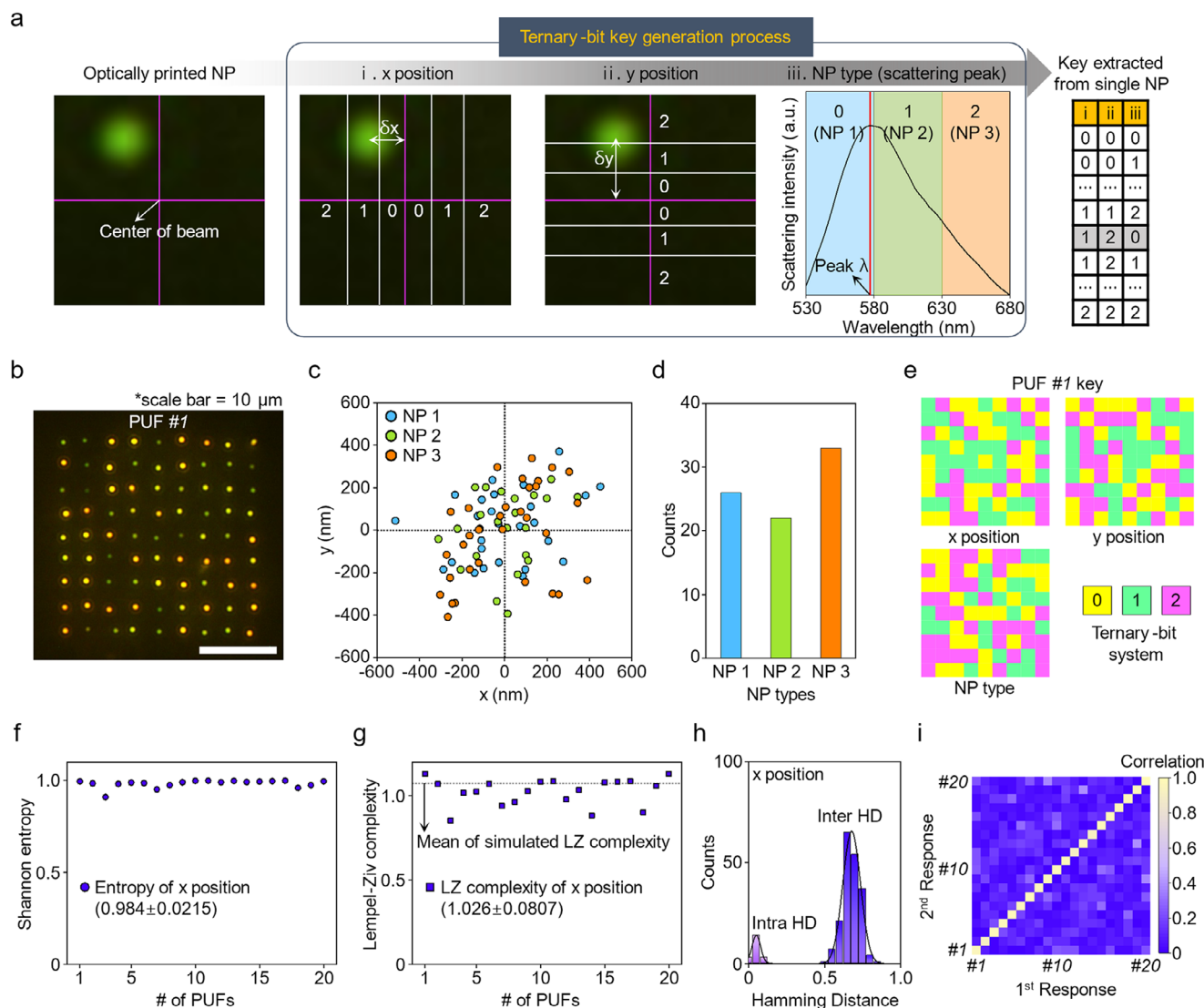


Figure 4. Key generation based on the random features and evaluation of the randomness of the keys. a) Schematic illustration of the PUF key generation process based on positional errors in the x and y-axis and NP type of a printed AuNP. b) Dark-field image of a PUF consisting of 81 AuNPs in a 9 $\times$ 9 lattice pattern. c) Center positions of 81 AuNPs in (b). NP 1, NP 2, and NP 3 denote 80 nm AuNP, 100 nm AuNP, and octahedral AuNP, respectively. d) Population of 81 AuNPs in (b). e) 2D images of ternary bit keys generated from (b) for display purposes. f) Shannon entropy of 20 PUF keys extracted from the x position. The ideal value of the entropy for true random keys is 1. g) Lempel–Ziv (LZ) complexity of 20 PUF keys extracted from x positions. The dotted line represents the mean value of 100 000 simulations of pseudo-random ternary keys with a sequence length of 81. h) Histogram of intra- and inter-hamming distance (HD) calculated among 20 PUF keys extracted from the x position. The black line shows the Gaussian fit of the distribution. The ideal value of inter HD for the ternary bit key is 0.6666. i) Heatmap representation of Pearson correlation coefficient among 20 PUF keys extracted from x position.

with a sequence length of 81 extracted based on x, y position of Figure 4c, and NP type of Figure 4d are visualized in Figure 4e as 9 $\times$ 9 2D color images for display purpose. After the key generation, diverse analyses were conducted on the extracted keys of 20 PUFs to assess the feasibility of utilizing x and y positions, and NP types in generating the unique PUF key.

The evaluation of the amount of information and complexity of PUF keys is essential for PUF. For true random keys in a ternary bit system, the probabilities of '0', '1', '2' in a bit string should be ideally equal to 0.333, corresponding to an amount of informa-

tion of 1. To evaluate the property, Shannon entropy calculation of the ternary keys was performed as below.^[83]

$$H_{\text{ternary}} = -(p_0 \log_3(p_0) + p_1 \log_3(p_1) + p_2 \log_3(p_2)) \quad (1)$$

where the probability (p_0, p_1, p_2) denotes the proportion of 0, 1, and 2 in the total number of bits from each PUF, and the sum of all values should be equal to 1. Using this equation, entropy values of the keys extracted from the x position y position, and NP type were calculated and presented in Figure 4f and Figure S13a (Supporting Information), indicating a uniform distribution of

'0', '1', '2'. However, since Shannon entropy provides only distribution information, we additionally utilized LZ complexity to assess the complexity of the keys (see Figure 4g; Figure S13b, Supporting Information). The LZ complexity values of the extracted keys are comparable to those obtained from 100 000 simulations of pseudo-random ternary keys of the same sequence length (dotted lines), indicating high complexity of the keys.

Additionally, the reproducibility and uniqueness of the PUF keys were assessed through Hamming distance (HD). Figure 4h and Figure S13c (Supporting Information) show the histogram of intra-HD and inter-HD. Intra-HD values were obtained from the same PUF but different trials (time points), while inter-HD values were obtained by comparing different PUFs, resulting in a total of 190 pairs. The values of intra HD and inter HD are 0.0512 ± 0.0295 and 0.6488 ± 0.0557 for the x position, 0.0667 ± 0.0338 and 0.6659 ± 0.0505 for the y position, and 0.0 ± 0.0 and 0.6638 ± 0.0544 for NP type, respectively. The value of 0 for the intra HD of NP type case can be attributed to the higher precision of the spectroscopic method compared to image-based measurements and clear criteria for NP type. These statistical values calculated across the three variables closely approximate the ideal values of 0.0 and 0.6667 for intra HD and inter HD,^[82,83] respectively, demonstrating reproducibility and uniqueness of the keys. Moreover, Pearson correlation of the keys across 20 PUFs was utilized for further assessment regarding reproducibility and uniqueness. The resulting correlation coefficients are visualized as the heatmap in Figure 4i and Figure S13d (Supporting Information). Inter- and intra-correlation coefficients are 0.0905 ± 0.0723 , 0.9615 ± 0.0234 for the x position, 0.0808 ± 0.0631 , 0.9490 ± 0.0263 for the y position, and 0.0866 ± 0.0638 , 1 ± 0.0 for NP type, respectively. Low inter-correlations (except diagonal in Figure 4i; Figure S13d, Supporting Information) and high intra-correlations (diagonal in Figure 4i; Figure S13d, Supporting Information) imply uniqueness and reproducibility of the keys.

Through comprehensive analyses, we could successfully demonstrate that the multi-modal ternary bit keys generated from x position, y position, and NP type exhibit uniformity, complexity, uniqueness, and reproducibility, which are the required properties for PUF. This represents the feasibility of exploiting the entire randomness, encompassing printed positions and NP types, occurring due to the Brownian motion during the optical printing of three types of AuNPs. Furthermore, this multi-modal system can ultimately enhance the security and applicability of PUFs.

2.4. Integration of Individual Keys into One Key

Given the independence of the three variables, evaluated through the mutual information between them (see Figure S14, Supporting Information), X , Y , and O denote x position, y position, and NP type, respectively), it is also possible to integrate the keys extracted from these variables to lengthen the bit string of the PUF key. Extending a sequence length in the key from 81 to 243 can potentially improve the capability of PUF. To validate this assumption, the three individual keys were integrated into one key, as illustrated in Figure 5a. In the integrated key, the keys for x position, y position, and NP type are placed at position

$3i - 2$, $3i - 1$, and $3i$ (where $i = 1, \dots, 81$), respectively. Consequently, a ternary bit key with a length of 3, comprising (x position, y position, NP type) in sequence, is generated from a single AuNP.

After the key integration, we performed the analyses on 20 integrated keys (see Figure S15, Supporting Information) to evaluate their capability of them. It is noteworthy that the counts of '0', '1', '2' in total integrated keys (243×20 PUFs = 4860) and NP types (81×20 PUFs = 1620) are nearly uniform (see Figure S16, Supporting Information), indicating the potential for longer bit strings to improve the uniformity and complexity. Accordingly, the Shannon entropy values of the integrated keys for each PUF increase and exhibit a more uniform distribution compared to the individual keys, as shown in Figure 5b. Likewise, the LZ complexity values also show a narrower distribution (see Figure 5c). Figure 5d shows the histogram of intra-HD and inter-HD calculated from the integrated keys. The intra- and inter-HD are 0.0393 ± 0.0153 and 0.6595 ± 0.0314 , respectively. These values are close to the theoretical ideal value of $2/3$ for inter-device HD in ternary bit system, which assumes a sequence with equal probabilities of 0, 1, and 2.^[82,83] The distribution is also narrower due to the larger number of bits per key, which reduces statistical variance in accordance with the Law of Large Numbers.^[89] Low inter-correlation and high intra-correlation were also observed, indicating the uniqueness and reproducibility of the integrated keys (see heatmap in Figure 5e). We also calculated the probability of cloning (PoC) using the intra- and inter-HD distributions to further assess the effectiveness of the integration strategy. As shown in Figure S17 (Supporting Information), the calculated PoC values across the independent variables and the integrated one quantitatively show that replication is highly impractical. Notably, the integrated key yields a PoC of 4.99×10^{-75} , reflecting the effectiveness of the integration approach for generating stronger PUFs.

Furthermore, we conducted a comprehensive modeling attack experiment to evaluate resistance of our ternary PUF system against machine learning-based modeling attacks.^[90] Four widely adopted machine learning algorithms were utilized for the experiment: logistic regression, support vector machine, multi-layer perceptron, and random forest. Assuming that the attacker has a partial knowledge of the ternary keys, each model was trained to predict unknown portions of the keys (see details in Note S2, Supporting Information). As shown in Table S1 (Supporting Information), prediction accuracy is concentrated near 0.33, with the highest value only slightly exceeding this baseline. Precision, recall, and F1 scores also remain in the low 0.3 range. Z-tests confirm that in all cases, the model performances are statistically indistinguishable from random guessing, with p -values consistently well above the 0.05 significance threshold. These results demonstrate that our PUFs exhibit strong resistance to modeling attacks, as even highly optimized machine learning models failed to uncover meaningful patterns in the key sequences.

The reproducibility and environmental robustness of PUFs are crucial for real-world applications. To evaluate the reproducibility and temporal stability of our PUFs, consecutive readout experiments on the same PUF device over short time intervals and measurements with different time points using different PUF devices were conducted, respectively. As shown in Figure S18a

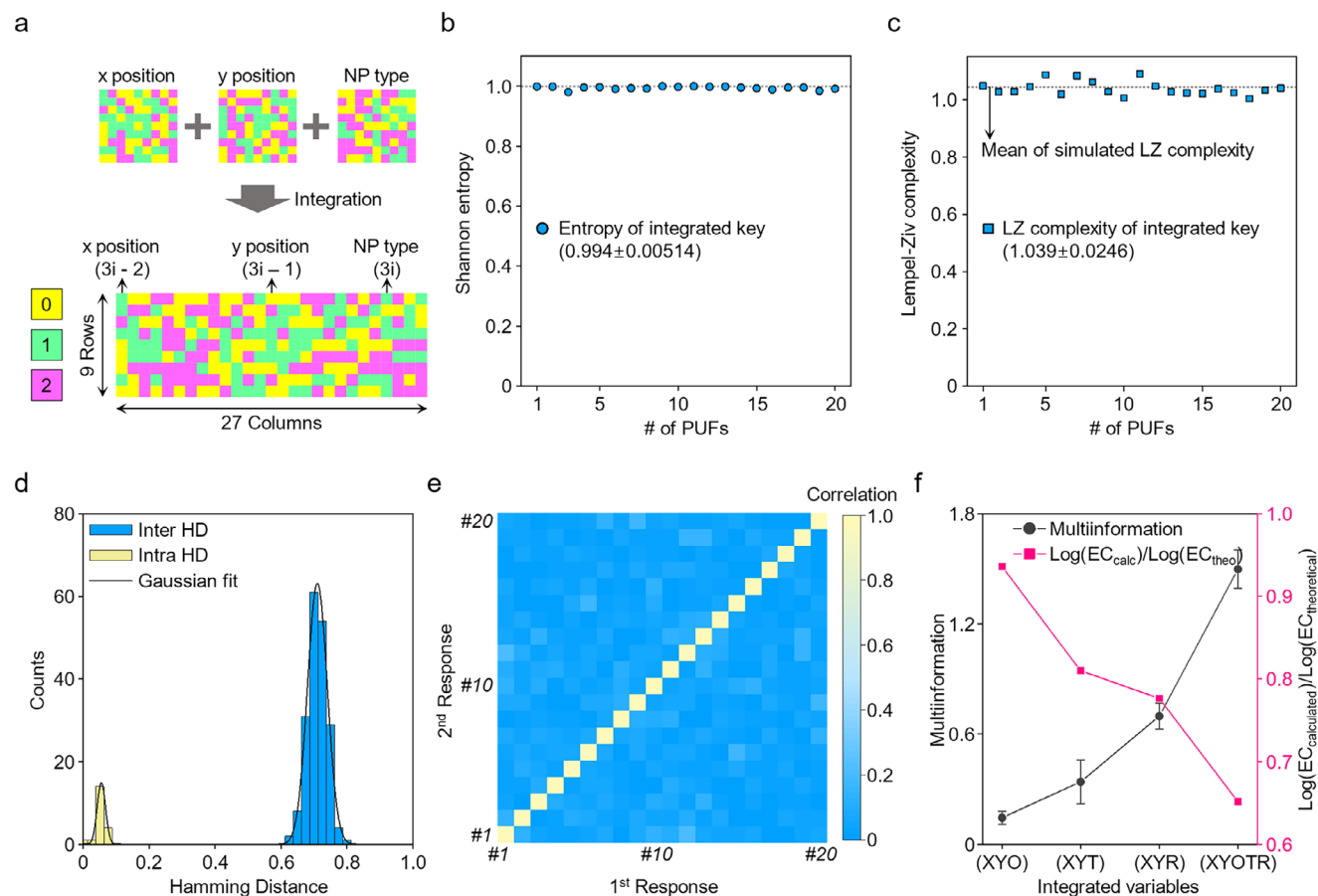


Figure 5. Integration of individual keys into one key and evaluation of the integrated keys. a) Schematic illustration of key integration. b) Shannon entropy and c) LZ complexity of 20 integrated keys. The dotted line in (c) represent the mean value of 100 000 simulations of pseudo-random ternary keys with a sequence length of 243. d) Histogram of intra- and inter HD calculated among 20 integrated keys. The black line shows the Gaussian fit of the distribution. e) Heatmap representation of Pearson correlation coefficient among 20 integrated keys. f) Relationship between multiinformation and encoding capacity. X, Y, O, T, and R denote the x position, y position, NP type, polar angle, and radial distance variables, respectively. (XYO), (XYT), (XYR), and (XYOTR) represent combinations of the variables for calculating multiinformation and encoding capacity.

(Supporting Information), the distribution of intra-HD values demonstrates consistent responses, indicating the reproducibility of our PUF device. In addition, as shown in Figure S18b (Supporting Information), the mean intra-HD values over time exhibit no significant monotonic trend. This is confirmed by a linear regression analysis, one-way analysis of variance (ANOVA), indicating that temporal fluctuations in the PUF responses are negligible over the period. This is further corroborated by the Spearman rank correlation coefficient between time and intra HD, supporting the absence of monotonic trends (see details in Figure S18a,b, Supporting Information). Importantly, all intra-HD values of those experiments remain below the authentication threshold calculated from Figure 5d, ensuring stable responses of the PUFs. We also assessed the environmental robustness of our PUFs by exposure of the PUFs to mechanical, thermal, and chemical perturbations (see details in Figure S18c, Supporting Information). In all tested environmental conditions, the intra-HD values remain within acceptable authentication thresholds, indicating that the PUF responses are not significantly changed by external perturbations. Although these experiments were performed in short-term periods, these results can provide evidence that the

reproducibility and environmental robustness of our PUFs meet the practical requirements for real-world implementation. To further improve the stability and robustness of the PUFs, passivation layers such as Al_2O_3 or polymer substances can be strategically implemented to protect PUFs from environmental perturbations or tampering attacks.

Additionally, we compared the encoding capacity between before and after integration. Theoretically, integration can enhance the encoding capacity from 3^{81} to 3^{243} . However, the actual encoding capacity based on the empirical results might not reflect this.^[25,81] In particular, if the individual keys are dependent, integrating them into a single key can increase the number of erroneous bits, such as false positives, thereby diminishing the significance of the theoretical encoding capacity. Therefore, we compared the actual encoding capacity before and after integration, which is derived from the degree of freedom.^[25,81,83,91] The degree of freedom for calculating the actual encoding capacity is as follows:

$$\text{Degree of Freedom} = \frac{p(1-p)}{\sigma^2} \quad (2)$$

where μ and σ denote the mean and standard deviation value of inter HD. According to this equation, the average value of the encoding capacity calculated from the three independent variables is $3^{78.6}$ ($\approx 10^{37}$). Integrating the variables increases the encoding capacity to $3^{227.4}$ ($\approx 10^{108}$), which represents a small relative error of 6.42% from the theoretical value. The relative error from the theoretical value, in comparison with the values of recently reported optical PUFs, is summarized in Table S2 (Supporting Information). To evaluate whether further integration can enhance PUF capability, we extracted keys from polar angle (T) and radial distance (R). Integrating keys from T and R into the integrated keys (XYO) increases the theoretical encoding capacity from 3^{243} to 3^{324} ($XYOT$ and $XYOR$) and 3^{405} ($XYOTR$) as the length of keys increases. However, the actual encoding capacities were observed as 3^{249} ($XYOT$), $3^{236.9}$ ($XYOR$), and $3^{263.9}$ ($XYOTR$), showing significant deviations from the theoretical capacities. This difference can be attributed to the dependency of R and T on X and Y variables. Despite having a narrower distribution due to the longer sequence lengths, mean values of inter HD of ($XYOT$), ($XYOR$), and ($XYOTR$) are found to deviate from the ideal value, indicating an increment of the false positive rate (see Figure S19, Supporting Information). To show the effect of dependence degree on encoding capacity, we calculated the dependence among variable combinations using multiinformation^[92] and plotted the relationship between multiinformation and encoding capacity in Figure 5f. Keys integrated with independent variables (XYO) demonstrate low multiinformation values and encoding capacity close to theoretical values. However, integrating additional dependent variables results in increased multiinformation values and deviates the encoding capacity further from theoretical values. This implies that when integrating individual keys extracted from various variables of a PUF, it is crucial to consider their independence. Otherwise, it may lead to merely an increase in the sequence length without significant improvement in performance.

Based on these analyses, we could prove that by integrating individual keys extracted from independent variables into a single key, thus extending the bit string, it is feasible to generate a stronger PUF even with a limited number of AuNPs.

2.5. Applicability of the Proposed PUF and Versatility of Optical Printing for Fabricating on-Demand PUF

In the previous section, we demonstrated an approach to enhance the capability of a PUF with a limited number of AuNPs (81 NPs) by integrating individual ternary bit keys extracted from independent variables into a single key. While increasing the number of AuNPs in a PUF would undoubtedly improve its capability, it is essential to figure out the extent to which the number of AuNPs can be decreased for fabricating on-demand PUFs within confined spaces like microelectronic devices.

Therefore, to determine the minimum number of NPs required for our proposed ternary PUF to function effectively, we cropped the center of the 9×9 pattern (81 NPs) to 7×7 (49 NPs), 5×5 (25 NPs), and 3×3 (9 NPs) patterns, as depicted in Figure 6a, to validate the capability of PUFs as the number of NPs decreases. Figure 6b shows the Shannon entropy corresponding to the number of NPs, revealing a decrease in entropy

with a broader distribution as the number decreases. Furthermore, the LZ complexity of the keys exhibits significant deviation for the case of 9 NPs (see Figure S20a, Supporting Information). This trend was corroborated by inter-HD values: for 49 NPs, it is 0.668 ± 0.0404 ; for 25 NPs, 0.664 ± 0.058 ; and for 9 NPs, 0.658 ± 0.0979 , indicating a broader distribution particularly evident with 9 NPs (see Figure 6c). Regarding statistical confidence concerns arising from sample size, we conducted additional simulations using a larger dataset to provide further support, as presented in Figure S21 (Supporting Information). A total of 1620 AuNPs indexed from 0 to 1619 were used to generate virtual key sequences. For each case, 200 sequences were generated by randomly selecting 9 NPs, 25 NPs, and 49 NPs from the database, and intra- and inter-HD were calculated accordingly. As shown in Figure S21 (Supporting Information), the histograms of simulated HD values are in strong agreement with those observed in the experimental data (Figure 6c), indicating that the sample size used in our study is statistically meaningful. As the sequence length of the keys reduced to 147, 75, and 27, the encoding capacities also decreased to $3^{136.3}$ ($\approx 10^{65}$), $3^{66.6}$ ($\approx 10^{31}$), and $3^{23.6}$ ($\approx 10^{11}$) for 49, 25, and 9 NPs, respectively (see Figure S20b, Supporting Information). Notably, PUFs made up of 25 NPs offer an encoding capacity surpassing $\approx 10^{20}$, effectively enhancing their resistance against counterfeiting. Considering these analysis results, we could assert that the proposed ternary PUF using as few as 25 AuNPs functions properly in terms of the amount of information, complexity, uniqueness, and encoding capacity.

The PUF should function as a key-based system; thus, we conducted challenge-response pair generation using a ternary-based exclusive OR (XOR) operation, accompanied by an external challenge generated by a random number generator (Figure 6d). The purpose of adopting an external challenge is to enlarge the challenge-response pair space and explore its potential applications, not only for anticounterfeiting but also in secure communications. The XOR operation is clearly defined in the binary system for digital logic and computer science. In a two-value input binary system, the definition of the XOR operation is that the output is true (1) only when the inputs are different (01 and 10); when the inputs are the same (00 and 11), the output will be false (0). Distinct from the binary system, the ternary system presents a unique challenge: with three possible inputs (0, 1, and 2), defining the XOR operation as simply '0 if identical, 1 if different' is insufficient, as it does not account for the third value, 2. Consequently, even though the inputs are 0, 1, and 2, the output may only be represented as 0 and 1 if defined strictly according to traditional XOR logic. To incorporate the third value, 2, as an output, it becomes necessary to introduce an alternative definition that expands the XOR operation to handle 0, 1, and 2 distinctly. This requires establishing new rules for interactions among all three values to produce outputs of 0, 1, or 2.^[93–95]

Therefore, we define a ternary XOR operation as $a \oplus b = (a + b) \bmod 3$, establishing a cycle-based truth table that ensures equal probabilities of 0, 1, and 2 as output results (see Figure S22a, Supporting Information). This approach allows for balanced outcomes and adapts the XOR logic to suit the increased complexity of a multinary system. In this XOR operation, the input numbers, generated by a random number generator, also maintain the same probability distribution, resulting in nearly ideal bit uniformity in the output.

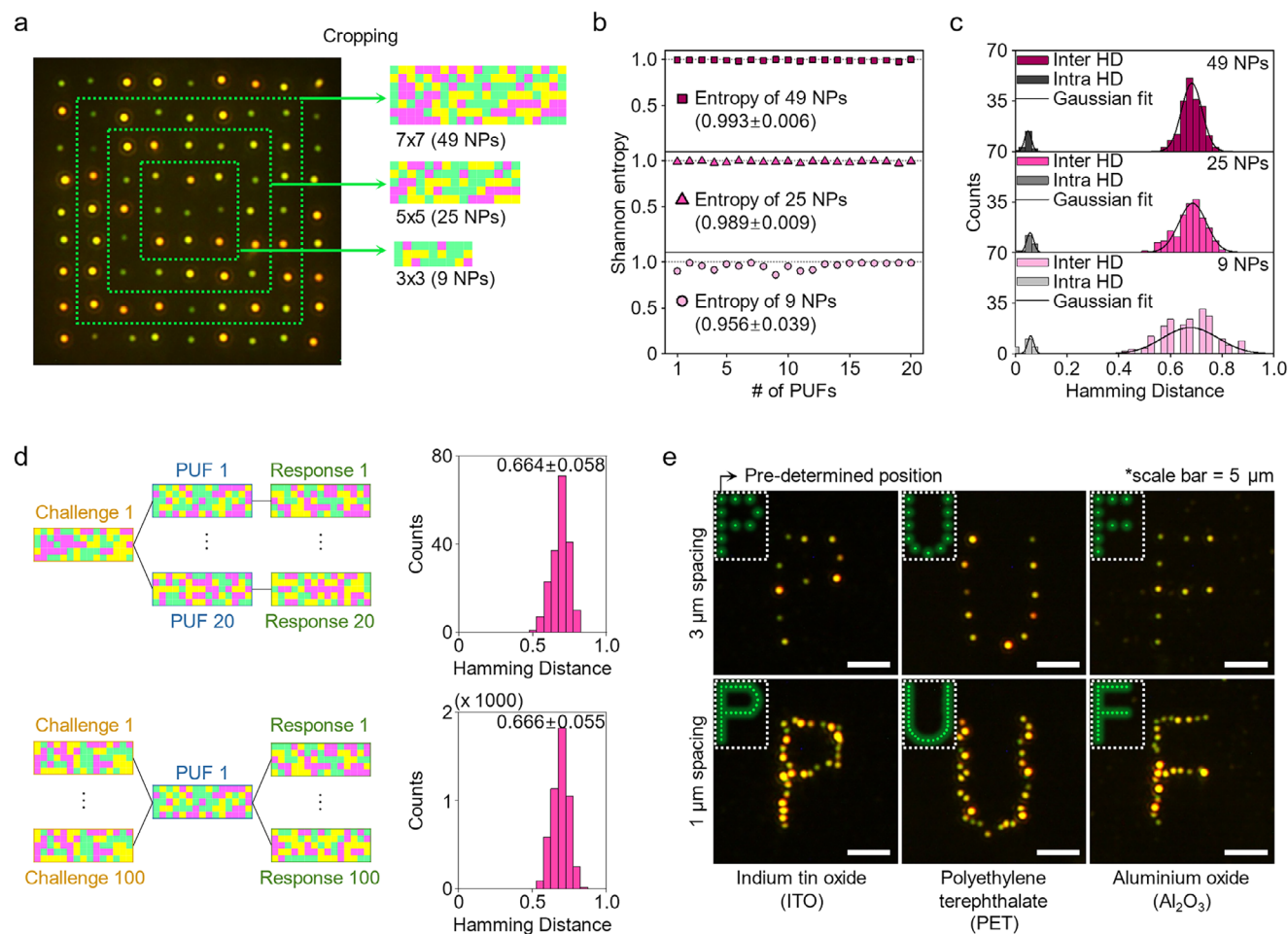


Figure 6. Demonstration of the applicability of our PUF and the versatility of the optical printing approach. a) Cropping process of a 9×9 patterns (81 AuNPs) to 7×7 (49 AuNPs), 5×5 (25 AuNPs), and 3×3 patterns (9 AuNPs) to determine the minimum number of AuNPs for our PUF. b) Shannon entropy of cropped PUFs. The dotted lines represent the ideal Shannon entropy value, which is 1 for true random keys. c) Histogram of intra- and inter-HD of cropped PUFs. d) Challenge-response pair generation through exclusive OR (XOR) operation, defined in a ternary system, using PUFs consisting of 25 AuNPs. The top represents one-challenge and multi-PUFs XOR and the bottom represents multi-challenges and one-PUF XOR. e) Dark-field images depict the letters 'P', 'U', 'F' as dots using three types AuNPs (80 nm AuNP, 100 nm AuNP, and octahedral AuNP). 'P', 'U', and 'F' are optically printed onto ITO, PET, and Al_2O_3 surfaces, respectively. Pre-determined spacing between NPs is set to 3 and 1 μm for upper and bottom images, respectively.

Figure 6d shows the results of the XOR operation using the PUF comprising 25 NPs. The top represents the case of the one-challenge and multi-PUFs to demonstrate whether the responses are identical or different when the same challenge is presented to different PUFs. The value of inter HD is 0.664 ± 0.058 which is close to the ideal value, indicating that the same challenge applied to different PUFs will yield distinguishable outputs. Conversely, the bottom shows the case of the multi-challenges and one-PUF. In a PUF comprising 25 NPs, the number of possible random challenges can be calculated as $75C_{25} \times 50C_{25} \times 25C_{25} \approx 6.65 \times 10^{33}$, indicating that the PUF can theoretically distinguish among 10^{33} different challenges with the same proportion of 0, 1, and 2. To demonstrate this, we performed the XOR operation with 100 random challenges. It was confirmed that the responses are distinct and distinguishable through inter-HD analysis, which is 0.666 ± 0.055 . Both XOR operations were also conducted with the same manner on the PUFs

comprising 49 NPs and 81 NPs, resulting in similar outcomes (see Figure S22b,c, Supporting Information). These statistical results prove that the proposed PUFs comprising several tens of NPs possess uniqueness. Furthermore, based on the implementation of ternary challenge-response pairs using structured XOR operations, our system provides not only stable and statistically uniform responses but also a flexible key format compatible with various authentication protocols. This includes lightweight authentication schemes such as COBRA,^[96] pairing frameworks like PARCE,^[97] and public PUF models such as PUFSec.^[98] These protocol architectures typically require reproducible and entropy-rich responses, which are satisfied by our system. Therefore, our optical PUF can be readily applied to real-world applications, such as cryptography^[25,91] and IoT protocols.^[99,100]

Despite the development of excellent PUFs, their practical use may be hindered without fabrication technology capable of creating PUFs in arbitrary sizes, configurations, and positions on

diverse materials. Taking this into account, optical printing can fundamentally offer remarkable fabrication flexibility. To demonstrate the capability of the optical printing approach for fabricating on-demand PUFs, we optically printed AuNPs with arbitrary configurations on various surfaces. Among numerous candidates of substrates, we adopted materials for demonstration purposes, including indium tin oxide (ITO), polyethylene terephthalate (PET), and aluminum oxide (Al_2O_3), since ITO is one of the most commonly used transparent conducting materials,^[101] PET is a widely used flexible material due to its mechanical and electrical insulating properties,^[102] and Al_2O_3 serves as an electrical insulator, high-k dielectric and passivation layer in micro-electronic devices.^[103] For instance, regarding relevance to semiconductor manufacturing such as CMOS microelectronics, the optical printing system can be applied in back-end-of-line (BEOL) or post-packaging stages because laser processes are frequently used in both stages and such ITO, SiO_2 , and Al_2O_3 surfaces are widely utilized for transparent electrodes in image sensors and display-integrated circuits and passivation layers or interlayer dielectrics in CMOS stacks, respectively. Additionally, the readout setup for our PUFs can be readily integrated into standard microelectronics fabrication processes. Due to their morphology-dependent distinct optical properties, the spectroscopic measurement can be replaced with a dark-field optical image-based measurement. Therefore, the image-based approach alone can enable efficient, straightforward, and scalable PUF readout without complex readout setup for practical purposes.

The optical dark-field images in Figure 6e illustrate the letters “P”, “U”, “F” represented as dots using AuNPs, printed on ITO, PET, and Al_2O_3 surfaces, respectively, and the insets denote pre-determined positions of the printing laser center. In the upper images of Figure 6e, the spacing between NPs was set to 3 μm , consistent with the PUF fabricated previously. If the spacing is set below 1 μm , the positions of printed NPs can possibly fall within the region of the focused laser beam used for subsequent printing events, as the spacing is comparable to the beam diameter. This may lead to light-matter interaction that triggers thermoplasmonic effects^[104] derived from the printed NPs and induces consequent thermophoresis and thermo-osmosis phenomena^[105,106] due to the temperature gradient generated near the surface of the NPs, potentially influencing subsequent printing events. Thus, to determine the feasibility of reducing the spacing, we optically printed NPs using the minimum achievable spacing in our setup, which is 1 μm , to avoid interference between printing events. As shown in the bottom images of Figure 6e, approximately three times more NPs were successfully printed in the same area size as the upper images. This signifies that more NPs can be printed in a confined space, ultimately improving the capability of the PUF. In other words, it allows for fabricating PUFs with the same capability in a smaller area. For instance, for PUFs consisting of 25 NPs in a 5×5 lattice pattern, reducing the spacing from 3 to 1 μm would allow for fabricating PUFs of equivalent performance within a smaller square area of $\approx 5 \mu\text{m} \times 5 \mu\text{m}$, indicating $\approx 85\%$ reduction in area. A comprehensive comparison with the current landscape of optical PUFs fabricated using various materials and methods, as presented in Table S2 (Supporting Information), highlights how this study fills previously unexplored areas of encoding density. It is notable that in the lattice pattern, the effective area occupied by 25

NPs is $\approx 0.25 \mu\text{m}^2$, which is not completely discernible to the eye. Furthermore, the spacing is not a fundamental limit of optical printing and can be further narrowed depending on factors such as the optical setup, type of NPs, and substrate. Consequently, through these demonstrations, we could explicitly manifest the versatility of optical printing, revealing its highly promising potential for fabricating on-demand PUFs within limited spaces on a broad range of substrates.

3. Conclusion

In summary, we have developed a plasmonic ternary PUF that exploits entire random characteristics induced by Brownian motion during the optical printing of AuNPs. The PUF consisting of several tens of AuNPs was optically printed using a mixture of three types of AuNPs as ink. The randomness of the features involving positional errors and types of printed AuNPs was comprehensively analyzed using statistical methods and information theory. Based on the random variables, three types of ternary bit keys with a sequence length equal to the number of AuNPs were generated and the individual keys consequently exhibited the ample capability required for PUF. To make stronger PUF in the limited number of AuNPs, we have suggested an approach that integrates the individual keys into a single key with considering the independence of the random variables and validated the improved capability of the integrated key in terms of the amount of information, complexity, uniqueness, and encoding capacity. Furthermore, we have figured out the minimum number of AuNPs, which is 25 NPs, required for the developed PUF to function effectively by systematically analyzing the capability for fabricating the PUF within confined spaces. We have further demonstrated the applicability of the developed PUF for applications like cryptography and IoT protocols through challenge-response pair generations using an exclusive OR operation defined in a ternary system. Last, we have shown the versatility of optical printing by demonstrating remarkable fabrication flexibility regarding the usability of a broad range of substrates and the ability to create arbitrary patterns with tunable dimensions. Table S2 (Supporting Information) positions our approach within the current framework of optical PUFs, underscoring its capabilities in encoding density and practical applicability.

This study mainly signifies two aspects. First, the Brownian motion of particles suspended in a fluid can be harnessed as the nature of randomness for fabricating PUFs. Although there is an ongoing debate whether the Brownian motion itself is truly random or not, we empirically demonstrated that the stochastic Brownian motion-to-deterministic optical forces transformation of our system induces both the physical features and the keys extracted based on these features exhibit randomness required for a unique and secure PUF. Second, the optical printing approach can serve as a pivotal method for directly fabricating additive optical PUFs. We utilized three types of AuNPs for the plasmonic ternary PUF, but there is no theoretical limit in the employment of the number and types of optical nanomaterials such as metallic NPs, quantum dots, upconversion nanocrystals, and 2D materials, implying the potential for fabrication of stronger multinary PUFs that can possess multi-modal challenge-response pairs. Furthermore, the outstanding versatility of optical printing offers a highly promising potential to directly fabricate on-demand

PUFs for a wide range of applications that require PUFs in confined spaces, such as microelectronic components of electronic devices.

4. Experimental Section

Materials: Poly(diallyldimethylammonium chloride) (PDAA, M_w 400 000–500 000, 20 wt.% in H_2O), gold(III) chloride trihydrate ($HAuCl_4$), 1,5-pentanediol were purchased from Aldrich and used without further purification. Polyethyleneterephthalate (PET) film with a thickness of 100 μm was purchased from Printec.

Synthesis of Gold Nanoparticles: Octahedral AuNP was synthesized following the previously reported method.^[107] PDAA solution (0.16 mL) and 4 μL of 2.5 M $HAuCl_4$ solution were sequentially added into 10 mL of pentanediol in a glass vial under vigorous stirring at room temperature. After stirring for 10 min, the vial was capped with a Teflon-liner cap, and the mixture was refluxed at 210 °C for 1 h in a pre-heated oil bath. After the reaction, the solution was centrifuged at 12 000 rpm for 15 min and the precipitate was redispersed with acetone once and with water twice.

Spherical AuNP was synthesized by etching the octahedral AuNP according to the previously reported method.^[108] A 1 μL of 2.5 M $HAuCl_4$ was added into the unwashed octahedral AuNP solution under vigorous stirring at room temperature. For 80 nm AuNP, octahedral AuNP synthesized at 220 °C was used. After the etching procedure for 20 h, the resultant AuNPs were collected by centrifugation at 12 000 rpm for 15 min and washed with acetone once and with water twice. The size of the synthesized AuNPs was determined using field emission scanning electron microscopy (JEOL, JSM7500F).

Fabrication of ITO and Al_2O_3 Films on Glass: A glass slide was cleaned successively with acetone, isopropyl alcohol, and water using an ultrasonic bath for 10 minutes each. After cleaning, 30 nm thick layers of ITO and Al_2O_3 were deposited on the glass substrate using atomic layer deposition at 200 °C and RF magnetron sputtering, respectively. The surfaces of ITO and Al_2O_3 were characterized by SEM-EDS analysis (see Figure S23, Supporting Information).

Optical Printing of AuNPs and Characterization: The optical setup for the optical printing of AuNPs is illustrated in Figure 1a. The setup was constructed on an inverted microscope (Olympus, IX73) equipped with a dark-field condenser (CytoViva). A 532 nm continuous wave diode-pumped solid-state laser (Pavilion Integration Corporation) was focused on the substrates through a beam expander and an oil-immersion 100 $\times$ objective lens (Olympus, UPLFLN100X, NA was set to 1) to optically print AuNPs suspended in water within a home-made chamber. Laser power was measured after passing through the beam expander. After printing an AuNP, the laser was immediately turned off using an optical shutter to avoid any unexpected printing until the next printing event. To implement the next printing event, the substrate was subsequently manipulated to a next position in designated patterns with a velocity of 1000 $\mu m s^{-1}$ using a piezo nanopositioning stage (Physik Instrumente), and the laser was simultaneously turned on. A CMOS camera (Thorlabs, DCC3260C) was used to visualize and record color images of AuNPs. To attain optical spectral responses of printed AuNPs, a spectrograph (Andor, Shamrock 303i) and an EMCCD (Andor, DU970P) were utilized.

Calculations of Statistical Analyses and Information Theory–K2 Function: K2 function $K_2(r)$,^[68] the first derivative of the pair correlation function $g(r)$, for a point pattern can be described as follows:

$$\hat{K}_2(r) = \frac{\hat{g}(r + \Delta r) - \hat{g}(r - \Delta r)}{2\Delta r} \quad (3)$$

where r is the distance from a randomly chosen point of the given pattern. Significant deviations of observed patterns were assessed from a null hypothesis of CSR (Complete Spatial Randomness) by using 999 Monte Carlo simulation envelopes. The observed values of $K_2(r)$ above or below the 25th highest and lowest simulation values at each scale r indicate regularity or aggregation of the given patterns, respectively. The calculations of

$K_2(r)$ and the simulation envelopes were performed using the Programita software package.^[109,110]

Mutual Information: The mutual information $MI(X; Y)$ of two discrete variables X and Y can be calculated as follows:

$$MI(X; Y) = \sum_{y \in Y} \sum_{x \in X} P_{(X,Y)}(x, y) \log \left(\frac{P_{(X,Y)}(x, y)}{P_X(x) P_Y(y)} \right) \quad (4)$$

where $P(x, y)$ is the joint probability of X and Y , and $P(x)$ and $P(y)$ are the marginal probabilities of X and Y , respectively.

For the estimation of mutual information of continuous variables in Figure 2g, the k -nearest neighbors (KNN) method was used with a parameter $k = 3$ implemented in the “Scikit-learn” library in Python according to previously reported studies.^[76]

Sample Entropy: For the given data set of the length of N , $X = [x_1, x_2, x_3, \dots, x_N]$ with a constant time interval τ , a template vector of embedding dimension m can be defined as follows:

$$X_m(i) = [x_i, x_{i+1}, x_{i+2}, \dots, x_{i+m-1}], i = 1, 2, \dots, N - m + 1 \quad (5)$$

Using $X_m(i)$ and the distance function $d[X_m(i), X_m(j)] = \max_{k=0, \dots, m-1} |x_{i+k} - x_{j+k}|$, where $i \neq j$ the probability of the number of possible template vector pairs can be defined as

$$C^m(r) = \frac{\text{number of pairs with } d[X_m(i), X_m(j)] \leq r}{(N - m + 1)(N - m)} \quad (6)$$

where r is the threshold distance. In this study, m is taken to be 2, and r is set as $r = 0.2\sigma$, where σ denotes the standard deviation of the overall data. Using $C^m(r)$, the sample entropy can be calculated as follows:

$$\text{Samp En}(m, r, N) = -\log \left(\frac{C^{m+1}(r)}{C^m(r)} \right) \quad (7)$$

Permutation Entropy: The permutation entropy of the given data set of the length of N , $X = [x_1, x_2, x_3, \dots, x_N]$ can be calculated by considering the relative occurrence frequency of possible ordinal patterns π . To calculate the ordinal pattern frequency, a sampling method can be applied using embedding dimension D and delay τ such that $X_i^{D,\tau} = [x_i, x_{i+\tau}, \dots, x_{i+(D-1)\tau}]$, where $i = 1, 2, \dots, N - (D - 1)\tau$. If $D = 3$, there are $3! = 6$ distinct ordinal patterns as follows: $\pi_1 = \{0, 1, 2\}$, $\pi_2 = \{0, 2, 1\}$, $\pi_3 = \{1, 0, 2\}$, $\pi_4 = \{1, 2, 0\}$, $\pi_5 = \{2, 0, 1\}$, $\pi_6 = \{2, 1, 0\}$. For each π_k , the probability $p^{D,\tau}(\pi_k)$ with each pattern frequency dividing by the total number of patterns can be estimated as follows:

$$p^{D,\tau}(\pi_k) = \frac{\# \{X_i^{D,\tau} | X_i^{D,\tau} \text{ has ordinal pattern } \pi_k\}}{N - (D - 1)\tau}, \text{ where } k = 1, \dots, D! \quad (8)$$

Using the probability $p^{D,\tau}(\pi_k)$, the permutation entropy can be defined by applying the definition of Shannon’s entropy as:

$$PE(D, \tau, N) = -\sum_{k=1}^{D!} p^{D,\tau}(\pi_k) \log(p^{D,\tau}(\pi_k)) \quad (9)$$

The permutation entropy PE can also be normalized such that

$$PE(D, \tau, N)_{\text{norm}} = -\sum_{k=1}^{D!} \frac{p^{D,\tau}(\pi_k) \log(p^{D,\tau}(\pi_k))}{\log D!} \quad (10)$$

In this study, embedding dimension $D = 3$ and delay $\tau = 1$ were chosen for the PE calculation.

Lempel–Ziv Complexity: A detailed calculation algorithm for Lempel–Ziv (LZ) complexity, C_{LZ} , can be found elsewhere.^[86] With this, the normalized LZ complexity S_{LZ} of the ternary bit key with the sequence length N can be calculated as

$$S_{LZ} = \frac{C_{LZ}}{\max(C_{LZ})} = -\frac{\log_3 N (P_0 \log_3 P_0 + P_1 \log_3 P_1 + P_2 \log_3 P_2)}{N} C_{LZ} \quad (11)$$

Hamming Distance: PUF uniqueness (inter-PUF Hamming distance (HD)) = $\frac{2}{q(q-1)} \sum_{i=1}^{q-1} \sum_{j=i+1}^q \frac{HD(K_i, K_j)}{s} \times 100\%$, where $HD(K_i, K_j)$ is the HD between K_i and K_j . This metric is essential for effective identification and authentication.

Readout reproducibility (intra-PUF HD) = $\frac{1}{m} \sum_{t=1}^m \frac{HD(K_i, K_{i,t})}{s} \times 100\%$, where K_i and $K_{i,t}$ are the original s -bit reference key and a s -bit key extracted from the same PUF at a different time-point t and m is the number of repeated measurements.

Multinformation: For a given set of n variables $\{X_1, X_2, \dots, X_n\}$, the multinformation $I(X_1, X_2, \dots, X_n)$ can be calculated as follows:

$$H(X_i) = -\sum_{x_i \in X_i} P(x_i) \log P(x_i) \quad (12)$$

$$H(X_1, X_2, \dots, X_n) = -\sum_{x_1 \in X_1} \sum_{x_2 \in X_2} \dots \sum_{x_n \in X_n} P(x_1, x_2, \dots, x_n) \log P(x_1, x_2, \dots, x_n) \quad (13)$$

$$I(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i) - H(X_1, X_2, \dots, X_n) \quad (14)$$

where $H(X_i)$ is the information entropy (Shannon entropy) of each variable X_i , and $H(X_1, X_2, \dots, X_n)$ is the joint entropy of n variables $\{X_1, X_2, \dots, X_n\}$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

Brownian motion, gold nanoparticles, optical printing, physical unclonable functions

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- [1] U. Guin, D. DiMase, M. Tehranipoor, *J. Electron. Test.* **2014**, 30, 9.
- [2] M. Tehranipoor, et al. *Counterfeit integrated circuits*, Springer, Berlin, Germany, **2015**.
- [3] U. Guin, K.e Huang, D. DiMase, J. M. Carulli, M. Tehranipoor, Y. Makris, *Proc. IEEE* **2014**, 102, 1207.
- [4] J. Stradley, M. Pecht, *Circuit World* **2012**, 38, 163.
- [5] N. Asadizanjani, et al. *Physical Assurance: For Electronic Devices and Systems*, Springer Nature, Berlin, Germany, **2021**, pp. 21–47.
- [6] K. Nyako, S. Devkota, F. Li, V. Borra, *Electronics* **2023**, 12, 4618.
- [7] A. Shamsoshoara, A. Korenda, F. Afghah, S. Zeadally, *Comput. Networks* **2020**, 183, 107593.
- [8] P. K. Sadhu, V. P. Yanambaka, A. Abdelgawad, *Sensors* **2022**, 22, 7433.
- [9] P. Kanaga Priya, R. Sivaranjani, K. Thangaraj, N. Alsharabi, in *Homomorphic Encryption for Financial Cryptography: Recent Inventions and Challenges*, Springer, Berlin, Germany, **2023**, pp. 221–258.
- [10] A. Mehmood, A. Shafique, M. Alawida, A. N. Khan, *IEEE Access* **2024**, 12, 27530.
- [11] E. Savaş, in *Proc. 6th Int. Conf. Security of Information and Networks*, Association for Computing Machinery, New York, NY, USA, **2013**, 7.
- [12] R. Pappu, B. Recht, J. Taylor, N. Gershenfeld, *Science* **2002**, 297, 2026.
- [13] R. Arppe, T. J. Sørensen, *Nat. Rev. Chem.* **2017**, 1, 0031.
- [14] Y. Gao, S. F. Al-Sarawi, D. Abbott, *Nat. Electron.* **2020**, 3, 81.
- [15] R. Maes, R. Maes, *Physically unclonable functions: Concept and constructions*, Springer, Berlin, Germany, **2013**.
- [16] A. Dodda, S. Subbulakshmi Radhakrishnan, T. F. Schranghamer, D. Buzzell, P. Sengupta, S. Das, *Nat. Electron.* **2021**, 4, 364.
- [17] U. Rührmair, et al., in *Proceedings of the 17th ACM conference on Computer and communications security*, Association for Computing Machinery, New York, **2010**, 237.
- [18] U. Rührmair, J. Solter, F. Sehnke, X. Xu, A. Mahmoud, V. Stoyanova, G. Dror, J. Schmidhuber, W. Burleson, S. Devadas, *IEEE Trans. Inf. Forensics Secur.* **2013**, 8, 1876.
- [19] W. Cao, H. Bu, M. Vinet, M. Cao, S. Takagi, S. Hwang, T. Ghani, K. Banerjee, *Nature* **2023**, 620, 501.
- [20] N. Kayaci, R. Ozdemir, M. Kalay, N. B. Kiremitler, H. Usta, M. S. Onses, *Adv. Funct. Mater.* **2022**, 32, 2108675.
- [21] J. Zhang, Y. Liu, C. Njel, S. Ronneberger, N. V. Tarakina, F. F. Loeffler, *Nat. Nanotechnol.* **2023**, 18, 1027.
- [22] Y. Liu, F. Han, F. Li, Y. Zhao, M. Chen, Z. Xu, X. Zheng, H. Hu, J. Yao, T. Guo, W. Lin, Y. Zheng, B. You, P. Liu, Y. Li, L. Qian, *Nat. Commun.* **2019**, 10, 2409.
- [23] F. Chen, Q. Li, M. o Li, F. Huang, H. Zhang, J. Kang, P. Wang, *Chem. Eng. J.* **2021**, 411, 128350.
- [24] H. Im, J. Yoon, J. Choi, J. Kim, S. Baek, D. H. Park, W. Park, S. Kim, *Adv. Mater.* **2021**, 33, 2102542.
- [25] H. Im, J. Yoon, B. So, J. Choi, D. H. Park, S. Kim, W. Park, *ACS Nano* **2024**, 18, 11703.
- [26] A. Wali, S. Das, *Adv. Mater.* **2023**, 35, 2205365.
- [27] A. F. Smith, P. Patton, S. E. Skrabalak, *Adv. Funct. Mater.* **2016**, 26, 1315.
- [28] Y. Zheng, C. Jiang, S. H. Ng, Y. Lu, F. Han, U. Bach, J. J. Gooding, *Adv. Mater.* **2016**, 28, 2330.

- [29] Q. Li, F. Chen, J. Kang, J. Su, F. Huang, P. Wang, X. Yang, Y. Hou, *Adv. Funct. Mater.* **2021**, 31, 2010537.
- [30] M. Ibrar, S. E. Skrabalak, *Small Struct.* **2021**, 2, 2100043.
- [31] S. A. Maier, *Plasmonics: fundamentals and applications*, Vol. 1, Springer, Berlin, Germany, **2007**.
- [32] P. Zijlstra, M. Orrit, *Rep. Prog. Phys.* **2011**, 74, 106401.
- [33] J. D. Smith, M. d. A. Reza, N. L. Smith, J. Gu, M. Ibrar, D. J. Crandall, S. E. Skrabalak, *ACS Nano* **2021**, 15, 2901.
- [34] H. Cheng, Y. Lu, D. Zhu, L. Rosa, F. Han, M. Ma, W. Su, P. S. Francis, Y. Zheng, *Nanoscale* **2020**, 12, 9471.
- [35] D. Yu, Z. Liu, C.-K. Hu, Y. Shen, Z.-J. Li, X.-X. Zhang, A.-G. Shen, *Chem. Eng. J.* **2023**, 475, 146063.
- [36] Z. Tang, X. Liu, X. Liu, J. Wu, W. Lin, X. Lin, G. Yi, *Adv. Eng. Mater.* **2022**, 24, 2101701.
- [37] M. J. Guffey, N. F. Scherer, *Nano Lett.* **2010**, 10, 4302.
- [38] A. S. Urban, A. A. Lutich, F. D. Stefani, J. Feldmann, *Nano Lett.* **2010**, 10, 4794.
- [39] J. Li, E. H. Hill, L. Lin, Y. Zheng, *ACS Nano* **2019**, 13, 3783.
- [40] J. n. Gargiulo, S. Cerrota, E. Cortés, I. L. Violi, F. D. Stefani, *Nano Lett.* **2016**, 16, 1224.
- [41] M. J. Guffey, R. L. Miller, S. K. Gray, N. F. Scherer, *Nano Lett.* **2011**, 11, 4058.
- [42] J. Do, M. Fedoruk, F. Jäckel, J. T. Feldmann, *Nano Lett.* **2013**, 13, 4164.
- [43] C. Zaza, I. L. Violi, J. Gargiulo, G. Chiarelli, L. Schumacher, J. Jakobi, J. Olmos-Trigo, E. Cortes, M. König, S. Barcikowski, S. Schlücker, J. J. Sáenz, S. A. Maier, F. D. Stefani, *ACS Photonics* **2019**, 6, 815.
- [44] M. A. Huerco, C. M. Maier, M. F. Castez, C. Vericat, S. Nedev, R. C. Salazar, A. S. Urban, J. Feldmann, *ACS Nano* **2016**, 10, 3614.
- [45] J. Do, K. N. Sediq, K. Deasy, D. M. Coles, J. Rodríguez-Fernández, J. Feldmann, D. G. Lidzey, *Adv. Opt. Mater.* **2013**, 1, 946.
- [46] M. Li, T. Lohmüller, J. Feldmann, *Nano Lett.* **2015**, 15, 770.
- [47] A. S. Urban, T. Pfeiffer, M. Fedoruk, A. A. Lutich, J. Feldmann, *ACS Nano* **2011**, 5, 3585.
- [48] S. Nedev, A. S. Urban, A. A. Lutich, J. Feldmann, *Nano Lett.* **2011**, 11, 5066.
- [49] Y. Bao, Z. Yan, N. F. Scherer, *J. Phys. Chem. C* **2014**, 118, 19315.
- [50] J. E. Melzer, E. McLeod, *Microsyst. Nanoeng.* **2021**, 7, 45.
- [51] A. Einstein, *Ann. Phys.* **1905**, 322, 549.
- [52] X. Bian, C. Kim, G. E. Karniadakis, *Soft Matter* **2016**, 12, 6331.
- [53] P. Langevin, *C. R. Acad. Sci.* **1908**, 146, 530.
- [54] T. Franosch, M. Grimm, M. Belushkin, F. M. Mor, G. Foffi, L. Forró, S. Jeney, *Nature* **2011**, 478, 85.
- [55] R. Huang, I. Chavez, K. M. Taute, B. Lukic, S. Jeney, M. G. Raizen, E.-L. Florin, *Nat. Phys.* **2011**, 7, 576.
- [56] T. Li, S. Kheifets, D. Medellin, M. G. Raizen, *Science* **2010**, 328, 1673.
- [57] A. Grigorenko, N. Roberts, M. Dickinson, Y. Zhang, *Nat. Photonics* **2008**, 2, 365.
- [58] M. Righini, A. S. Zelenina, C. Girard, R. Quidant, *Nat. Phys.* **2007**, 3, 477.
- [59] Y. Zhang, C. Min, X. Dou, X. Wang, H. P. Urbach, M. G. Somekh, X. Yuan, *Light: Sci. Appl.* **2021**, 10, 59.
- [60] M. C. Wu, *Nat. Photonics* **2011**, 5, 322.
- [61] L. Lin, E. H. Hill, X. Peng, Y. Zheng, *Acc. Chem. Res.* **2018**, 51, 1465.
- [62] M. Braun, F. Cichos, *ACS Nano* **2013**, 7, 11200.
- [63] L. Lin, M. Wang, X. Peng, E. N. Lissek, Z. Mao, L. Scarabelli, E. Adkins, S. Coskun, H. E. Unalan, B. A. Korgel, L. M. Liz-Marzán, E.-L. Florin, Y. Zheng, *Nat. Photonics* **2018**, 12, 195.
- [64] A. Lehmuskero, P. Johansson, H. Rubinsztein-Dunlop, L. Tong, M. Kall, *ACS Nano* **2015**, 9, 3453.
- [65] A. S. Urban, S. Carretero-Palacios, A. A. Lutich, T. Lohmüller, J. Feldmann, F. Jäckel, *Nanoscale* **2014**, 6, 4458.
- [66] J. Gargiulo, I. L. Violi, S. Cerrota, L. S. Chvátal, E. Cortés, E. M. Perassi, F. Diaz, P. Zemanek, F. D. Stefani, *ACS Nano* **2017**, 11, 9678.
- [67] I. L. Violi, L. P. Martinez, M. Barella, C. Zaza, L. Chvátal, P. Zemanek, M. V. Gutierrez, M. Y. Paredes, A. F. Scarpettini, J. Olmos-Trigo, V. R. Pais, I. D. Noblega, E. Cortes, J. J. Saenz, A. V. Bragas, J. Gargiulo, F. D. Stefani, *J. Chem. Phys.* **2022**, 156, 034201.
- [68] K. Schiffrers, F. M. Schurr, K. Tielbörger, C. Urbach, K. Moloney, F. Jeltsch, *Ecography* **2008**, 31, 545.
- [69] B. D. Ripley, *J. Appl. Probab.* **1976**, 13, 255.
- [70] B. D. Ripley, *J. R. Stat. Soc.: Ser. B* **1977**, 39, 172.
- [71] B. D. Ripley, *J. R. Stat. Soc.: Ser. B* **1979**, 41, 368.
- [72] B. D. Ripley, *Spatial statistics*, John Wiley & Sons, Hoboken, NJ, USA, **2005**.
- [73] M. Ben-Said, *Ecol. Process.* **2021**, 10, 1.
- [74] Y. Zhu, S. Getzin, T. Wiegand, H. Ren, K. Ma, *PLoS One* **2013**, 8, 74560.
- [75] A. Kraskov, H. Stögbauer, P. Grassberger, *Phys. Rev. E* **2004**, 69, 066138.
- [76] B. C. Ross, *PLoS One* **2014**, 9, 87357.
- [77] A. Delgado-Bonal, A. Marshak, *Entropy* **2019**, 21, 541.
- [78] C. Bandt, B. Pompe, *Phys. Rev. Lett.* **2002**, 88, 174102.
- [79] J. H. Kim, S. Jeon, J. H. In, S. Nam, H. M. Jin, K. H. Han, G. G. Yang, H. J. Choi, K. M. Kim, J. Shin, S.-W. Son, S. J. Kwon, B. H. Kim, S. O. Kim, *Nat. Electron.* **2022**, 5, 433.
- [80] T. Zhang, L. Wang, J. Wang, Z. Wang, M. Gupta, X. Guo, Y. e. Zhu, Y. C. Yiu, T. K. C. Hui, Y. Zhou, C. Li, D. Lei, K. H. Li, X. Wang, Q. i. Wang, L. Shao, Z. Chu, *Nat. Commun.* **2023**, 14, 2507.
- [81] J. W. Leem, M. S. Kim, S. H. o Choi, S.-R. Kim, S.-W. Kim, Y. M. Song, R. J. Young, Y. L. Kim, *Nat. Commun.* **2020**, 11, 328.
- [82] Z. Hu, J. M. M. L. Comeras, H. Park, J. Tang, A. Afzali, G. S. Tulevski, J. B. Hannon, M. Liehr, S.-J. Han, *Nat. Nanotechnol.* **2016**, 11, 559.
- [83] S. Gandla, C. Moon, S. Baek, H. Park, S. Kim, *Adv. Funct. Mater.* **2023**, 33, 2211762.
- [84] G. Park, Y. S. Choi, S. J. Kwon, D. K. Yoon, *Adv. Mater.* **2023**, 35, 2303077.
- [85] T. M. Squires, T. G. Mason, *Annu. Rev. Fluid Mech.* **2010**, 42, 413.
- [86] A. Lempel, J. Ziv, *IEEE Trans. Inf. Theory* **1976**, 22, 75.
- [87] Y. Cho, J. Pyeon, H. Jang, G. Mun, J. Kang, B.-G. Park, G. Y. Kim, H. Kim, Y. S. Jung, *Adv. Funct. Mater.* **2025**, 2424079.
- [88] Y. Gu, C. He, Y. Zhang, L. i. Lin, B. D. Thackray, J. Ye, *Nat. Commun.* **2020**, 11, 516.
- [89] K. Yao, J. Gao, *IEEE Trans. Fuzzy Syst.* **2015**, 24, 615.
- [90] N. Hassan, U. Chatterjee, in *2024 IEEE Int. Symp. Hardware Oriented Security and Trust (HOST)*, IEEE, New York, NY, USA, **2024**, 361–372.
- [91] S. Gandla, J. Yoon, C.-W. Yang, H. Lee, W. Park, S. Kim, *Nat. Commun.* **2024**, 15, 7592.
- [92] M. Studený, J. Vejnárova, *Learning in Graphical Models*, Springer, Dordrecht, Netherlands, **1998**, 261.
- [93] Hurst, *IEEE Trans. Comput.* **1984**, C-33, 1160.
- [94] S. Lin, Y.-B. Kim, F. Lombardi, *IEEE Trans. Nanotechnol.* **2009**, 10, 217.
- [95] M. P. Singh, J. K. Rakshit, M. Hossain, *Opt. Quantum Electron.* **2021**, 53, 703.
- [96] J. Plusquellic, M. Areno, *Cryptography* **2018**, 2, 21.
- [97] D. Heeger, J. Plusquellic, in *16th Int. Conf. Distributed Computing in Sensor Systems (DCOSS)*, IEEE, New York, NY, USA, **2020**, 458–465.
- [98] S.-Y. Park, et al. in *IEEE INFOCOM 2017-IEEE Conf. Computer Communications.*, IEEE, New York, NY, USA, **2017**, 1–9.
- [99] A. Al-Meer, S. Al-Kuwari, *ACM Comput. Surv.* **2023**, 55, 1.
- [100] W. Liu, L. Zhang, Z. Zhang, C. Gu, C. Wang, M. O'Neill, F. Lombardi, *ACM Trans. Embedded Comput. Syst.* **2019**, 18, 1.
- [101] K. Ellmer, *Nat. Photonics* **2012**, 6, 809.
- [102] H. Xiang, Z. Li, H. Liu, T. Chen, H. Zhou, W. Huang, *npj Flexible Electron.* **2022**, 6, 15.
- [103] A. Sen, J. Shim, A. Bala, H. Park, S. Kim, *Adv. Funct. Mater.* **2023**, 33, 2301919.

- [104] L. Jaffred, A. Samadi, H. Klingberg, P. M. Bendix, L. B. Oddershede, *Chem. Rev.* **2019**, *119*, 8087.
- [105] C. Hong, S. Yang, J. C. Ndukaife, *Nat. Nanotechnol.* **2020**, *15*, 908.
- [106] J. Gargiulo, T. Brick, I. L. Violi, F. C. Herrera, T. Shibanuma, P. Albella, F. G. Requejo, E. Cortés, S. A. Maier, F. D. Stefani, *Nano Lett.* **2017**, *17*, 5747.
- [107] Y. Lu, H. Zhang, F. Wu, H. Liu, J. Fang, *RSC Adv.* **2017**, *7*, 18601.
- [108] Y.-J. Lee, N. B. Schade, L. i Sun, J. A. Fan, D. R. i Bae, M. M. Mariscal, G. Lee, F. Capasso, S. Sacanna, V. N. Manoharan, G. I-R. A Yi, *ACS Nano* **2013**, *7*, 11064.
- [109] T. Wiegand, A. Moloney, *Oikos* **2004**, *104*, 209.
- [110] T. Wiegand, K. A. Moloney, *Handbook of Spatial Point-Pattern Analysis in Ecology*, CRC Press, New York, **2013**.